

PROGRAM ASSESSMENT REPORT

v2024.1

Tracking data

CIM code	BS-AMAT-2
Program name	Bachelor of Science in Applied Mathematics
year	AY 24/25
Program assessment coordinator	Kiah Wah Ong
Assessment plan year	AY 24/25
No assessment this year per plan	☐
LO(s) assessed	LO 5 see below for description.
Presented at program authority meeting date: <i>Please attach minutes</i>	

As outlined in the program assessment document, we will evaluate the following program goal for AY 24-25:

LO 5: Communicate clearly and precisely, orally and in writing, their understanding of mathematical concepts.

Methodology

Three questions, Problem 1, 2, and 11, were selected from the final exam of MATH 402: Complex Analysis to assess Learning Objective 5. Each question is worth 5 points and provides students with an opportunity to communicate clearly and precisely in writing their understanding of the mathematical concepts learned in this course. Below are the assessment questions.

- Assessment question

Problem 1. (5 points)

Find the image S' of S under the given complex mapping

$$f(z) = \frac{1}{z}$$

if $S = \{x + iy \in \mathbb{C} : x = 1\}$

Note: Please explain your reasoning and provide a clear sketch of S' .

Problem 2. (5 points)

Show that

$$f(z) = f(x + iy) = x^2 - x + y + i(y^2 - 5y - x)$$

is not analytic at any point but is differentiable along $y = x + 2$.

Problem 11. (5 points)

Given

$$f(z) = \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\sin(z^2)}$$

show that $z = 0$ is a removable singularity and supply a definition of $f(0)$ so that f is analytic at $z = 0$.

- Rubrics for each question

Scores	Description	Indicators of Achievement
5	Demonstrates a complete and clear understanding of the concept.	Solution is mathematically correct, logically structured, and fully justified. Explanations are precise, coherent, and use appropriate mathematical language and notation. Communication shows mastery of both computation and conceptual understanding.
4	Shows solid understanding with minor errors or omissions.	Reasoning is mostly correct and well explained. Small computational or notational errors that do not affect overall validity. Clear communication of ideas, though could be more polished or precise.
3	Demonstrates partial understanding of the concept.	Correct approach but contains one or more significant computational or logical errors. Explanation lacks clarity or completeness in parts. Communication of reasoning is understandable but somewhat disorganized.

2	Shows limited understanding or significant misconceptions.	Partial or incorrect method with some relevant ideas. Explanations are unclear, incomplete, or mostly incorrect. Use of mathematical language is inconsistent or imprecise.
1	Demonstrates minimal understanding.	Response is largely incorrect or irrelevant. Explanations are missing or fail to convey understanding. Mathematical communication is unclear or incorrect throughout.
0	No meaningful attempt to solve or explain.	Blank, irrelevant, or incoherent response.

- Using the rubric above, we apply the following criteria to determine how well the learning objective has been achieved.

0 – 5 points	Does not meet expectations
6 – 10 points	Meets expectations
11 – 15 points	Proficient

0 – 5 points: Does Not Meet Expectations

The student shows limited understanding of the mathematical concepts and communicates ideas ineffectively. Explanations may be unclear, incomplete, or incorrect, with inconsistent use of mathematical language and notation. Written or oral responses lack organization and do not convey logical reasoning.

6 – 10 points: Meets Expectations

The student demonstrates an adequate understanding of the concepts and communicates reasoning with general clarity. Solutions are mostly correct and explanations are understandable, though they may contain minor errors, lack precision, or need stronger justification. Communication is coherent but could be more polished and rigorous.

11 – 15 points: Proficient

The student demonstrates a thorough understanding of mathematical concepts and communicates ideas clearly, precisely, and logically. Explanations are well-organized, fully justified, and use appropriate mathematical language and notation. Written and oral communication reflect both conceptual mastery and clarity of thought.

- Using three questions instead of a single question provides a broader and more reliable assessment of students' ability to communicate mathematical understanding clearly and precisely. Each question can target different concepts, problem types, or levels of reasoning within Complex Analysis, allowing students to demonstrate their communication skills across multiple contexts. This approach has several advantages:
 - **Improved Reliability:**

Assessing the same learning objective through multiple questions reduces the influence of chance, topic familiarity, or misunderstanding of a single problem. It gives a more stable and representative measure of each student's ability.
 - **Broader Coverage:**

Multiple questions allow the evaluation of both computational explanations and conceptual or proof-based reasoning, reflecting the variety of communication demands in advanced mathematics.
 - **Deeper Insight into Learning:**

Students may communicate clearly in one context but struggle in another. Using three questions helps identify specific strengths and weaknesses in both understanding and written communication.
 - **Fairer Evaluation:**

Multiple data points provide a more balanced and equitable assessment of the learning objective, minimizing the impact of one poor response or misinterpretation.

Results

- The table below displays the number and percentage of students at each level of proficiency.

Points	Assessments Categories	Number of students	Percentage of students in each category.
0 – 5 points	Does not meet expectations	0	0.0 %
6 – 10 points	Meets expectations	2	40%
11 – 15 points	Proficient	3	60%

- Because of the small sample size, we include all 5 of the scanned artifacts here.

- Artifact 1: The student obtains 9 out of 15 points (Student's A number: A20529983)

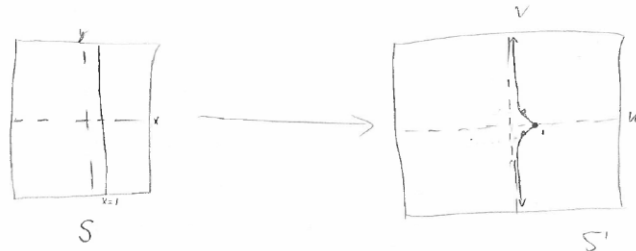
Problem 1. (5 points)

Find the image S' of S under the given complex mapping

$$f(z) = \frac{1}{z}$$

$$\text{if } S = \{x + iy \in \mathbb{C} : x = 1\}$$

Note: Please explain your reasoning and provide a clear sketch of S' .



$z = 1 + iy$
 $f(z) = \frac{1}{z} = \frac{1}{1 + iy}$

$$= \frac{1}{1 + iy} \cdot \frac{1 - iy}{1 - iy} = \frac{1 - iy}{1 + y^2} = \frac{1}{1 + y^2} + i \frac{-y}{1 + y^2}$$

$u(x, y) = \frac{1}{1 + y^2}$
 $v(x, y) = \frac{-y}{1 + y^2}$

y	u	v
0	1	0
1	1/2	-1/2
-1	1/2	1/2
10	1/101	-10/101
$y \rightarrow \infty$	$u \rightarrow 0$	$v \rightarrow 0$

Problem 2. (5 points)

Show that

$$f(z) = f(x + iy) = x^2 - x + y + i(y^2 - 5y - x)$$

is not analytic at any point but is differentiable along $y = x + 2$.

Cauchy
 Riemann
 for Analyticity

$u_x = v_y$
 $u_y = -v_x$

$u(x, y) = (x^2 - x + y)$
 $v(x, y) = (y^2 - 5y - x)$

$u_x = 2x - 1$
 $u_y = 1$
 $v_x = -1$
 $v_y = 2y - 5$

$u_x = v_y \Rightarrow 2x - 1 = 2y - 5$
 $2x + 4 = 2y$
 $y = x + 2$

$u_y = -v_x \Rightarrow 1 = -(-1)$
 $1 = 1$

True for all values (x, y)

Analytic
 IFF

Problem 11. (5 points)

Given

$$f(z) = \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\sin(z^2)}$$

show that $z = 0$ is a removable singularity and supply a definition of $f(0)$ so that f is analytic at $z = 0$.

$$f(z) = \frac{1}{\sin(z^2)} - \frac{\frac{1}{2}z^{10}}{\sin(z^2)} - \frac{\cos(z^5)}{\sin(z^2)}$$

$$= \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\sin(z^2)}$$

$$= \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\left[(z^2) - \frac{(z^2)^3}{3!} + \frac{(z^2)^5}{5!} - \dots \right]}$$

Laurent Series

$$\frac{1}{z^2} + 3z^5 + \dots$$

$$\left[z^2 - \frac{(z^2)^3}{3!} + \frac{(z^2)^5}{5!} - \dots \right] \cdot \left[\frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{z^2 - \frac{(z^2)^3}{3!} + \frac{(z^2)^5}{5!} - \dots} \right]$$

$$+ \frac{z^{10}}{2}$$

This is a removable singularity, as we only have a ^{single} pole of order 2.

If we define $f(0) = 1$, $\lim_{z \rightarrow 0} \left(\frac{1}{z^2} \left(\frac{1}{z^2} + 3z^5 - \dots \right) \right)$

this makes $f(z)$ analytic at $z = 0$.

1	3	/
2	5	/
3	3	/
4	2	/
5	1	/
6	5	/
7	5	/
8	4	/
9	3	/
10	0	/
11	1	/
12	5	/
13	1	/

- Artifact 2: The student obtains 10 out of 15 points (Student's A number: A20529769)

✓ Problem 1. (5 points)

Find the image S' of S under the given complex mapping

$$f(z) = \frac{1}{z}$$

$$\text{if } S = \{x + iy \in \mathbb{C} : x = 1\}$$

Note: Please explain your reasoning and provide a clear sketch of S' .

$$z = x + iy; \text{ real } x = 1$$

$$f(z) = \frac{1}{z} = \frac{1}{x + iy} \cdot \frac{x - iy}{x - iy} = \frac{x - iy}{x^2 + y^2}$$

$$\text{Real } \frac{x}{x^2 + y^2}$$

$$\text{Imaginary } \frac{-y}{x^2 + y^2}$$

$$u = \frac{x}{x^2 + y^2}$$

$$v = \frac{-y}{x^2 + y^2}$$

$$u = \frac{x}{x^2 + y^2} \Rightarrow x = u(x^2 + y^2) \Rightarrow x^2 + y^2 = \frac{x}{u}$$

$$x = \frac{x}{x^2 + y^2} \Rightarrow x^2 + y^2 = 1$$

$$u = \frac{x}{x^2 + y^2} = \frac{x}{1} = x$$

$$u x^2 + u y^2 = x \Rightarrow u x^2 - x = -u y^2$$

$$u = \frac{x}{x^2 + y^2}$$

$$u(x^2 + y^2) = x$$

Answer
On Back
or Exam

$$u x(x - 1) = -u y^2$$

$$x(x - 1) = -y^2$$

$$f(z) = \frac{1}{z} \quad z = x + iy \quad w = u + iv \quad x = 1$$

$$\frac{1}{x + iy} \cdot \frac{x - iy}{x - iy} = \frac{x - iy}{x^2 + y^2} = \frac{x}{x^2 + y^2} - \frac{iy}{x^2 + y^2}$$

$$u = \frac{x}{x^2 + y^2}$$

$$v = \frac{-y}{x^2 + y^2}$$

$$\Rightarrow \frac{-y}{1 + y^2} = v$$

$$u = \frac{1}{1 + y^2}$$

$$u(1 + y^2) = 1 \Rightarrow 1 + y^2 = \frac{1}{u} \Rightarrow y^2 = \frac{1}{u} - 1$$

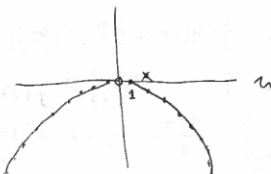
$$y = \pm \sqrt{\frac{1}{u} - 1} \quad \text{Plug in ...}$$

$$v = \frac{\pm \sqrt{\frac{1}{u} - 1}}{1 + \frac{1}{u} - 1} = \frac{\pm \sqrt{\frac{1}{u} - 1}}{\frac{1}{u}} = \pm u \sqrt{\frac{1}{u} - 1}$$

$$v = \pm u \sqrt{\frac{1}{u} - 1}$$

$$y = -x \sqrt{\frac{1}{x} - 1}$$

$$\begin{aligned} \text{if } |v| &= 1 \\ &= -1\sqrt{1-1} \\ &= -1\sqrt{0} \\ &= 0 \end{aligned}$$



$$\begin{aligned} u &= 2 \\ v &= -2\sqrt{\frac{1}{2} - 1} \\ v &= -2\sqrt{0} \end{aligned}$$

1	3
2	5
3	3
4	3
5	3
6	3
7	2
8	1
9	5
10	3
11	2
12	4
13	3

Problem 2. (5 points)

Show that

$$f(z) = f(x+iy) = \sqrt{x^2 - x + y} + i\sqrt{y^2 - 5y - x}$$

is not analytic at any point but is differentiable along $y = x + 2$.

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} ; \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial x} = 2x - 1 \quad \frac{\partial u}{\partial y} = 2y - 5$$

$$\frac{\partial v}{\partial x} = -1 \quad \frac{\partial v}{\partial y} = 1$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$2x - 1 = 2y - 5$$

$$1 = -(-1)$$

$$1 = 1 \text{ True for all } x, y$$

$$2x = 2y - 4$$

$$x = y - 2$$

$$-y - 2$$

$$-x - 2$$

$$y = x + 2$$

It is not analytic at any point since in $y = 1$ and $x = 1$ the, would not be analytic

However using Cauchy-Riemann equations we can say that it is differentiable along $y = x + 2$

However using Cauchy-Riemann equations we can say that it is differentiable along $y = x + 2$

that it is differentiable along $y = x + 2$

$$\sin(z) = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots$$

Problem 11. (5 points)

Given

$$f(z) = \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\sin(z^2)}$$

show that $z = 0$ is a removable singularity and supply a definition of $f(0)$ so that f is analytic at $z = 0$.

When $z = 0$ $\sin(z) = 0 \dots$ DNE in this question...

$$f(z) = \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\sin(z^2)} \quad \sin(z^2) = z^2 - \frac{z^4}{3!} + \frac{z^6}{5!} - \frac{z^8}{7!} + \dots$$

$$1 - \frac{1}{2}z^{10} - \cos(z^5) \left[\frac{1}{z^2} - \frac{3!}{z^4} + \frac{5!}{z^6} - \frac{7!}{z^8} + \dots \right]$$

$$1 - \frac{1}{2}z^{10} - \left[1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots \right] \left[\frac{1}{z^2} - \frac{3!}{z^4} + \frac{5!}{z^6} - \frac{7!}{z^8} + \dots \right]$$

If you multiply out the function you

will get a singularity at $\frac{1}{z^2}$.

to much more to do...

$$\cos(z^5) = 1 - \frac{z^{10}}{2!} + \frac{z^{20}}{4!} - \dots \quad \cos(z^2) = 1 - \frac{z^4}{2!} + \frac{z^8}{4!} - \dots$$

$$\cos(z^5) = 1 - \frac{z^{10}}{2!} + \frac{z^{20}}{4!} - \dots$$

- Artifact 3: The student obtains 13 out of 15 points (Student's A number: A20480073)

Problem 1. (5 points)

Find the image S' of S under the given complex mapping

$$f(z) = \frac{1}{z}$$

if $S = \{x + iy \in \mathbb{C} : x = 1\}$

Note: Please explain your reasoning and provide a clear sketch of S' .

$$f(z) = \frac{1}{z} = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2}$$

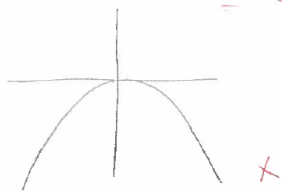
Set $x = 1$:

$$f(z) = \frac{1-iy}{1+y^2}$$

$$u(x, y) = \frac{1}{1+y^2}$$

$$v(x, y) = \frac{-iy}{1+y^2}$$

$$v = -iy \cdot u \Rightarrow u = -\frac{v}{iy}$$



Problem 2. (5 points)

Show that

$$f(z) = f(x + iy) = x^2 - x + y + i(y^2 - 5y - x)$$

is not analytic at any point but is differentiable along $y = x + 2$.

$$f(z) = x^2 - x + y + i(y^2 - 5y - x)$$

$$u(x, y) = x^2 - x + y, \quad v(x, y) = y^2 - 5y - x$$

$$u_x = 2x - 1, \quad v_x = -1$$

$$u_y = 1, \quad v_y = 2y - 5$$

Notice $u_y = -v_x$ but $u_x \neq v_y$, by theorem $f(z)$ is not analytic

If $y = x + 2$,

$$v_y = 2(x+2) - 5 = 2x + 4 - 5 = 2x - 1 = u_x$$

therefore $f(z)$ is differentiable when $y = x + 2$

Problem 11. (5 points)

Given

$$f(z) = \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\sin(z^2)}$$

show that $z = 0$ is a removable singularity and supply a definition of $f(0)$ so that f is analytic at $z = 0$.

$$\cos(z^5) = 1 - \frac{(z^5)^2}{2!} + \frac{(z^5)^4}{4!} - \frac{(z^5)^6}{6!} + \frac{(z^5)^8}{8!} - \dots$$

$$\sin(z^2) = z^2 - \frac{(z^2)^3}{3!} + \frac{(z^2)^5}{5!} - \dots$$

$$f(z) = \frac{1 - \frac{1}{2}z^{10} - (1 - \frac{1}{2}z^{10} + \frac{1}{4!}z^{20} - \dots)}{z^2 - \frac{1}{3!}z^6 + \frac{1}{5!}z^{10} - \dots}$$

$$= \frac{\cancel{z^2} [\frac{1}{4!}z^{18} - \frac{1}{6!}z^{28} + \frac{1}{8!}z^{38} - \dots]}{z^2 [1 - \frac{1}{3!}z^4 + \frac{1}{5!}z^8 - \dots]}$$

$$f(z) = \frac{\frac{1}{4!}z^{18} - \frac{1}{6!}z^{28} + \frac{1}{8!}z^{38} - \dots}{1 - \frac{1}{3!}z^4 + \frac{1}{5!}z^8 - \dots}$$

$$f(0) = \frac{\frac{1}{4!} \cdot 0^{18} - \frac{1}{6!} \cdot 0^{28} + \frac{1}{8!} \cdot 0^{38} - \dots}{1 - \frac{1}{3!} \cdot 0^4 + \frac{1}{5!} \cdot 0^8 - \dots} = \frac{0}{1} = 0$$

$\Rightarrow f$ is analytic at 0 and $z=0$ is a removable singularity

1	3 ✓
2	5 ✓
3	0
4	2 ✓
5	5 ✓
6	3 ✓
7	3 ✓
8	0
9	5 ✓
10	1 ✓
11	5 ✓
12	5 ✓
13	3 ✓

- Artifact 4: The student obtains 14 out of 15 points (Student's A number: A20484532)

Problem 1. (5 points)

Find the image S' of S under the given complex mapping

$$\left(\frac{1}{2} - u\right)^2 = \frac{1}{4} + 2\left(\frac{1}{2}\right)u - u^2$$

$$f(z) = \frac{1}{z}$$

if $S = \{x + iy \in \mathbb{C} : x = 1\}$

Note: Please explain your reasoning and provide a clear sketch of S' .

$$\text{let } z = x + iy \rightarrow \frac{1}{z} = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}$$

sub $x=1$:

$$\frac{1 - iy}{1 + y^2} = u + iv \Rightarrow \frac{1}{1 + y^2} = u, \quad \frac{-y}{1 + y^2} = v$$

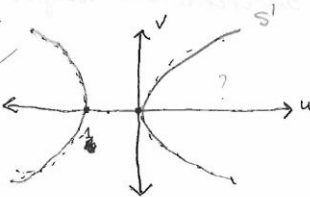
$$\therefore \sqrt{\frac{1}{u} - 1} = y \quad [(u \neq 0) \text{ not possible}]$$

$$\rightarrow \frac{\pm \sqrt{\frac{1}{u} - 1}}{1 + (\frac{1}{u} - 1)} = v$$

$$\rightarrow \pm u \sqrt{\frac{1}{u} - 1} = v \rightarrow u^2 \left(\frac{1}{u} - 1\right) = v^2$$

$$\rightarrow u - u^2 = v^2 \quad \left(\frac{1}{2} - u\right)^2 - \frac{1}{4} = v^2$$

$$= \left(u + \frac{1}{2}\right)^2 - v^2 = \left(\frac{1}{2}\right)^2$$



Problem 2. (5 points)

Show that

$$f(z) = f(x + iy) = x^2 - x + y + i(y^2 - 5y - x)$$

is not analytic at any point but is differentiable along $y = x + 2$.

$$u = x^2 - x + y$$

$$v = y^2 - 5y - x$$

$$u_x = 2x - 1$$

$$v_y = 2y - 5$$

$$u_y = 1$$

$$v_x = -1$$

$$u_y = -v_x \quad \checkmark$$

$$u_x = v_y \Rightarrow 2x - 1 = 2y - 5$$

$$\therefore 2x + 4 = 2y$$

$$\therefore y = x + 2$$

$f(z)$ continuous and

$$(u_y = -v_x \wedge u_x = v_y) \text{ only for } y = x + 2$$

$\Rightarrow f(z)$ differentiable only for $y = x + 2$.

not analytic anywhere since DNE
a differentiable neighborhood.

Problem 11. (5 points)
Given

$$f(z) = \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\sin(z^2)}$$

$$\frac{5z^9 - 5z^4 \sin(z^5)}{2z \cos(z^2)}$$

show that $z = 0$ is a removable singularity and supply a definition of $f(0)$ so that f is analytic at $z = 0$.

$$\sin(z^2) = z^2 - \frac{z^6}{3!} + \frac{z^{10}}{5!} - \dots$$

zero of order 2

$$\cos(z^5) = 1 - \frac{z^{10}}{2!} + \frac{z^{20}}{4!} - \dots$$

zero of order 20

$$\Rightarrow 1 - \frac{1}{2}z^{10} - \cos(z^5) = \frac{z^{20}}{4!} - \frac{z^{30}}{6!} + \dots$$

order 20

$\therefore z = 0$ is a zero of order $20 - 2 = 18$ for $f(z)$

$\therefore z = 0$ is a removable singularity.

$$\lim_{z \rightarrow 0} \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\sin(z^2)} \xrightarrow{L'H} \lim_{z \rightarrow 0} \frac{5z^9 - 5z^4 \sin(z^5)}{2z \cos(z^2)}$$

$$= \lim_{z \rightarrow 0} \frac{5z^8 - 5z^3 \sin(z^5)}{2 \cos(z^2)} = 0$$

define $f(0) = 0$

1	4
2	5
3	5
4	5
5	5
6	5
7	5
8	2
9	4
10	5
11	5
12	5
13	3

- Artifact 5: The student obtains 14 out of 15 points (Student's A number: A20554585)

Problem 1. (5 points)

Find the image S' of S under the given complex mapping

$$f(z) = \frac{1}{z} \quad \text{circle}$$

if $S = \{x + iy \in \mathbb{C} : x = 1\}$

Note: Please explain your reasoning and provide a clear sketch of S' .

$$z = x + iy, \quad f(z) = \frac{1}{z} = \frac{1}{x + iy} = \frac{1}{\cos \theta + i \sin \theta} = \frac{1}{e^{i\theta}}$$

$$\text{real: } x = 1 \quad \text{imaginary: } \left\{ \begin{array}{l} u = \frac{x}{x^2 + y^2} \\ v = \frac{-y}{x^2 + y^2} \end{array} \right.$$

$$\Rightarrow \frac{x}{x^2 + y^2} = 1 \quad \frac{-y}{x^2 + y^2} = 0$$

$$u = \frac{x}{x^2 + y^2} \Rightarrow (x^2 + y^2)u = x$$

$$v(x^2 + y^2) = -y$$

$$\left\{ \begin{array}{l} \frac{-y}{v} = (x^2 + y^2) \\ \frac{x}{u} = (x^2 + y^2) \end{array} \right. \quad \frac{-y}{v} = \frac{x}{u} \Rightarrow \frac{1}{v} = \frac{-x}{uy} \Rightarrow v = \frac{uy}{-x}$$

$$u = \frac{x}{x^2 + y^2} \quad v = \frac{-y}{x^2 + y^2}$$

$$\Rightarrow u = \frac{1}{1 + y^2}, \quad v = \frac{-y}{1 + y^2}$$

$$\Rightarrow -y = uv$$

Problem 2. (5 points)

Show that

$$f(z) = f(x + iy) = x^2 - x + y + i(y^2 - 5y - x)$$

is not analytic at any point but is differentiable along $y = x + 2$.

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial x} = 2x - 1, \quad \frac{\partial v}{\partial y} = 2y - 5, \quad \frac{\partial u}{\partial y} = -1, \quad \frac{\partial v}{\partial x} = -1$$

$$2x - 1 = 2y - 5 \quad \text{when } y = x + 2 \Rightarrow 2x - 1 = 2(x + 2) - 5$$

$$\Rightarrow 2x - 1 = 2x - 1$$

$f(z)$ is only differentiable when $y = x + 2$

$\therefore f(z)$ has no holomorphic regions defined

$\Rightarrow f(z)$ is not analytic $\forall z \in \mathbb{C}$

due to satisfaction of
Cauchy-Riemann eq.

$$\lim_{z \rightarrow z_0} f(z) = L$$

Problem 11. (5 points)
Given

removable singularity

$$f(z) = \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\sin(z^2)}$$

show that $z = 0$ is a removable singularity and supply a definition of $f(0)$ so that f is analytic at $z = 0$.

$$\begin{aligned} \lim_{z \rightarrow 0} \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\sin(z^2)} &= \lim_{z \rightarrow 0} \frac{-\frac{10}{2}z^9 + \sin(z^5) \cdot 5z^4}{2z \cos(z^2)} \\ &= \lim_{z \rightarrow 0} \frac{-5z^9 + 5z^4 \sin(z^5)}{2z \cos(z^2) + 2z \sin(z^2)} = 0 \end{aligned}$$

$\therefore z = 0$ is a removable singularity

$f(0) = 0$ to be analytic

$$f(z) = \begin{cases} \frac{1 - \frac{1}{2}z^{10} - \cos(z^5)}{\sin(z^2)} & z \neq 0 \\ 0 & z = 0 \end{cases}$$

analytic $\therefore$

1	4
2	5
3	5
4	3
5	3
6	4
7	4
8	2
9	5
10	0
11	5
12	4
13	3

Continuous Improvement, CI, plans

- The assessment data provide meaningful insights into students' performance on Learning Objective 5: Communicate clearly and precisely, orally and in writing, their understanding of mathematical concepts.

Findings:

All students successfully met this learning objective. Their written work on the selected exam questions demonstrated clear reasoning, logical organization, and appropriate use of mathematical language and notation. Students effectively expressed their understanding of key concepts in Complex Analysis through well-structured explanations and justifications.

Oral communication component:

At present, the oral communication aspect of this learning objective is not formally scored. Instead, it is observed informally through class participation, such as discussions, problem presentations, and engagement during lectures. This informal observation allows instructors to gauge how students explain their reasoning verbally and respond to questions in real time. In future iterations of the course, a formal assessment component—such as short student presentations or oral problem explanations—will be introduced to more systematically evaluate and document students' oral communication skills.

Plan for Improvement:

While we are pleased with these strong results, we are committed to continuous improvement. To further strengthen students' communication and critical thinking skills, we plan to implement the following strategies:

1. Advanced problem-solving and proof communication:

Introduce more challenging problems and proof-based questions that require students to write detailed explanations of their reasoning. This will help students refine their ability to express complex mathematical ideas clearly and precisely.

2. Formal oral assessments:

Incorporate structured oral presentations or problem explanations to formally assess students' verbal communication of mathematical ideas. This will provide a more balanced evaluation of both written and oral components of Learning Objective 5.

3. Ongoing feedback and reflection:

Continue collecting feedback from students on both written and oral communication activities. Regular reflection opportunities will help students become more aware of how effectively they convey mathematical reasoning and where they can improve.

By focusing on these initiatives, we aim to maintain our high success rate while deepening students' ability to communicate mathematical ideas clearly, logically, and confidently in both written and oral forms.